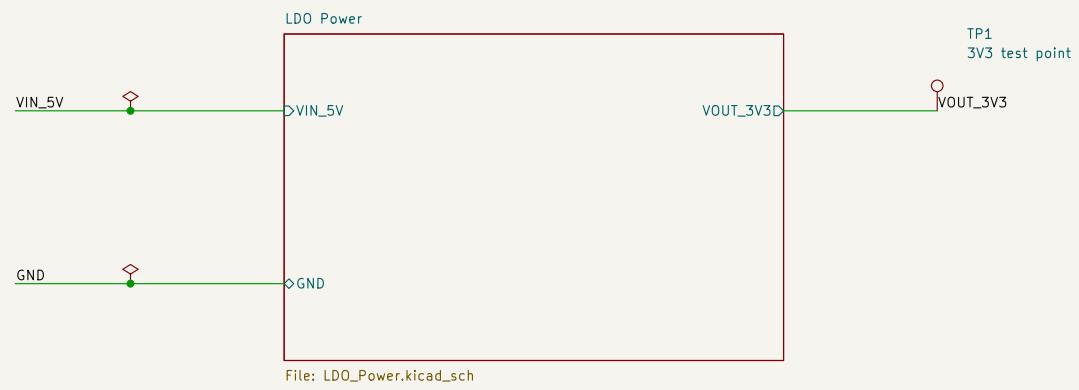


LDO integration example



VIN_5V: from 5 V buck supply
VOUT_3V3: to STM32 and BNO085 IMU power
GND: common system ground

Open the LDO Power sheet to edit the regulator circuit.
Import LDO_Power.kicad_sch into the system as an existing hierarchical sheet.
See README.md for library transfer and integration steps.

PWR_FLAG symbols here describe the external source for standalone ERC.
The reusable child sheet deliberately has no global rail symbols or source flags.

5 V to 3.3 V, 1 A total rail capacity
Senior Design

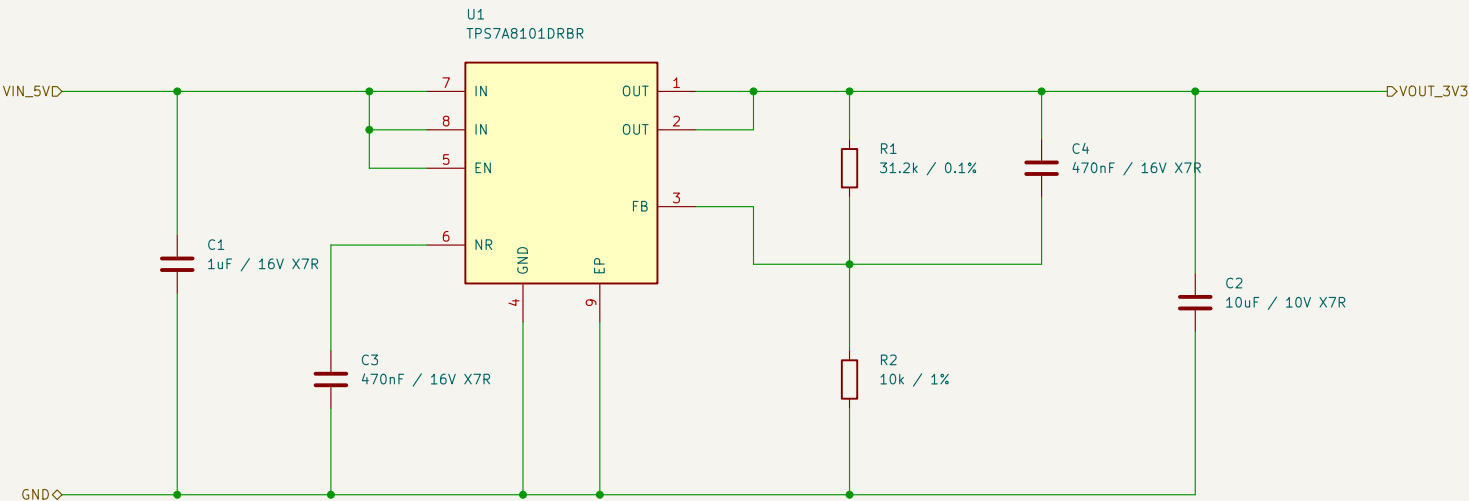
Sheet: /
File: STM32_LDO.kicad_sch

Title: LDO supply – integration example

Size: A4	Date: 2026–10–05	Rev: 1.0
KiCad E.D.A. 10.0.6		Id: 1/2

5 V to 3.3 V LDO supply

Reusable sheet: VIN_5V input, VOUT_3V3 output, common GND



VOUT nominal = $0.8 \times (1 + 31.2k / 10k) = 3.296 \text{ V}$
1 A is the TOTAL output budget for STM32 + IMU + other 3.3 V loads.

C2: require $\geq 4.7\mu\text{F}$ EFFECTIVE capacitance; selected part modeled at 8.8uF.
EN tied to IN. C3 sets noise reduction / soft start; C4 bypasses R1.

Layout: capacitors close to U1; EP pad 9 to GND copper and thermal vias.
Heat at 1 A: about 1.7 W at 5 V input; verify junction temperature on final PCB.

5 V motor / encoder supply branches BEFORE this sheet.
Adafruit BNO085 breakout: 3.3 V rail to VIN; 3Vo is an output, leave separate.
Bare BNO085: connect 3.3 V to VDD and VDDIO with its specified decoupling.

5 V to 3.3 V, 1 A total rail capacity

Senior Design

Sheet: /LDO Power/

File: LDO_Power.kicad_sch

Title: LDO power supply – reusable sheet

Size: A4

Date: 2026–10–05

Rev: 1.0

KiCad E.D.A. 10.0.6

Id: 2/2